Sean Luka Girgis

Senior Data Engineer | Python, SQL, PySpark, AWS & AI-Ready Data Pipelines

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, AWS data platforms, Redshift, ETL/ELT pipelines, data warehousing, data quality, documentation, mentoring, telemetry analytics, and AI/ML-ready data workflows. Strong background building scalable data pipelines, curated reporting layers, validation checks, performance-tuned SQL, and operational documentation for regulated enterprise environments. Best aligned to senior data engineering roles requiring Spark/PySpark, SQL, Python, cloud data warehousing, ETL/ELT, data quality, collaboration, documentation, and AI/RAG-supporting data foundations; healthcare EHR/EMR, HL7/FHIR, Kafka/Flink, NiFi/Informatica/Fivetran, and LangChain/LlamaIndex are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, analytics, and downstream reporting workloads.
- Developed data-centric reporting and analytics applications using Python, SQL, Streamlit, matplotlib, seaborn, and plotly to surface capacity, performance, and trend insights for senior stakeholders.
- Prepared curated, model-ready datasets for Prophet and scikit-learn forecasting workflows, supporting AI/ML-adjacent capacity planning and bottleneck prediction.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve completeness, consistency, and trust in pipeline outputs.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Documented data flows, metric definitions, transformation rules, operational runbooks, support steps, and design notes to improve repeatability, source control discipline, and knowledge transfer.
- Mentored and supported engineers and stakeholders by explaining pipeline behavior, troubleshooting production data issues, and creating reusable patterns for reporting, validation, and operational analysis.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.

- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported AWS cloud migration planning by evaluating monitoring readiness, transaction visibility, and data/reporting needs.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and reporting processes for telecom-scale transaction systems.

- Worked with XML, SQL, Oracle, PL/SQL, messaging, and integration interfaces supporting billing, customer-management, and operational reporting workflows.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built automation scripts in Korn shell, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine
Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

AI-Powered Job Search Pipeline
Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.
Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, RAG-style Retrieval, Quality Gates